

Sahaj Raj Malla

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Computational mathematics undergraduate researcher in deep learning, quantum machine learning, and multi-agent systems. Author of MallaNet (*Scientific Reports*, 2025) and eight further peer-reviewed papers and preprints spanning computer vision, time-series forecasting, and game-playing agents. Current direction: reinforcement learning for multi-modal agentic systems.

Education

B.Sc. in Computational Mathematics

Dec 2022 – Dec 2026 (expected)

Kathmandu University, Dhulikhel, Nepal

Selected Training & Fellowships

Aspire Leaders Program, Aspire Institute (Harvard University)	Jan 2025 – Mar 2025
AI Fellowship 2023, Fusemachines Nepal	Jan 2023 – Jan 2024
Womanium Quantum 2023, Womanium & QWorld	Jul 2023 – Aug 2023
Data Science Fellowship with DataCamp, Code for Nepal	Dec 2021 – Aug 2023
Annual Nepal AI School (2023, 2024 cohorts), NAAMII	May 2023, Dec 2024
Introduction to Quantum Computing, The Coding School & IBM	Sep 2021 – Apr 2022

Publications

Peer-Reviewed Publications

- S. R. Malla. MallaNet: Residual Branch Merge Convolutional Neural Network with Homogeneous Filter Capsules for Devanagari Character Recognition.** *Scientific Reports (Springer Nature)*, Dec 2025.
99.71% test accuracy on the Devanagari Handwritten Character Dataset, 56.41% fewer parameters than prior SOTA; introduces optimized residual blocks and Homogeneous Filter Capsule layers for multiscale feature extraction.
[Single Author | Q1 Journal]
- S. R. Malla.** A landmark-based addressing framework for urban navigation using geospatial clustering and pathfinding algorithm. *Journal of Science, Engineering and Technology (JSET)*, 19(1), Mar 2025.
37.7% lower average path length and 22.9% lower travel time vs. current systems, using DBSCAN clustering, H3 indexing, and A* search over OpenStreetMap data.
- B. Kafle, S. R. Malla, A. Shrestha, S. Jaiswal.** Evaluating XGBoost Algorithm for Nepali Stock Price Prediction. *Int. Conf. on Technologies for Computer, Electrical, Electronics & Communication (IEEE)*, Oct 2023.
XGBoost-based prediction model for the Nepali stock market.

Preprints

4. **S. R. Malla.** ArthaVyu: A Stacking Ensemble Framework for Multi-Source Real Estate Price Prediction in Nepal. *Research Square*, Jun 2026.
Stacking ensemble framework for multi-source real estate price prediction in Nepal; integrates property listings, geospatial data, and market indicators for improved valuation accuracy.
5. **S. R. Malla.** Kalimati Vegetable Price Index Forecasting with a Momentum Corrected Online Stacking Ensemble. *arXiv:2605.30720*, May 2026.
Built the KVPI, an inverse-volatility-weighted index over 135 daily wholesale commodities (Kathmandu, 2013–2023); tested 14 forecasting models across 7/14/30/90-day horizons. Proposed ensemble reached RMSE 1.771, MAPE 0.68% at 90 days, explaining 84.5% of variance.
6. **S. R. Malla.** External Demand, Domestic Monetary Conditions, and Remittance Dynamics in Nepal. *arXiv:2605.19401*, May 2026.
ARDL-ECM model of remittances as a share of Nepal's GDP (1993–2024) using PCA-derived demand/monetary indices; error-correction term corrects ~26% of disequilibria annually, projects remittances at ~28.3% of GDP by 2030 under baseline conditions.
7. **S. R. Malla.** XGBoost Forecasting of NEPSE Index Log Returns with Walk-Forward Validation. *arXiv:2601.08896*, Jan 2026.
Walk-forward validation framework for time-series forecasting of NEPSE index log returns with XGBoost.
8. **P. Panth, S. R. Malla.** Secure, Verifiable, and Scalable Multi-Client Data Sharing via Consensus-Based Privacy-Preserving Data Distribution. *arXiv:2601.00418*, Jan 2026.
Consensus-driven privacy-preserving framework for secure, verifiable multi-client data sharing.
9. **S. R. Malla.** AI Agents for the Dhumbal Card Game: A Comparative Study. *arXiv:2510.11736*, Oct 2025.
Compared AI strategies for the Dhumbal card game; an aggressive heuristic agent reached an 88.3% win rate across 1024 simulations.
10. **S. R. Malla.** Devanagari Digit Recognition using Quantum Machine Learning: A Hybrid Quantum-Classical Approach. *arXiv:2506.09069*, Jun 2025.
First hybrid quantum-classical architecture for Devanagari digit recognition (CNN + 10-qubit variational circuit); 99.80% test accuracy, 0.9980 average per-class F1.

Professional Experience

Founder

Nov 2023 – Present

HamroNiwas, Banepa, Nepal

- Built a rental and marketplace platform for Nepali students, connecting 4,000+ users and reducing search time by 90%.

Data Consultant

May 2025 – Jun 2025

YoungInnovations Pvt. Ltd., Kathmandu, Nepal

- Designed and optimized ELT pipelines with Apache Spark for large-scale dataset processing; automated workflows with Apache Airflow. (Contract.)

Data Scientist Sep 2021 – Jan 2023

Naamche Technology, Kathmandu, Nepal

- Scraped and analyzed 1M+ real estate listings (Florida, Texas, California); produced investment-decision reports, improving decision accuracy by 40+% and reducing computation time by 200+%.

Data Engineer Mar 2021 – Sep 2021

Devfinity Nepal, Kathmandu, Nepal

- Built a Django-based data factory processing 1M+ job records daily, deployed on Azure.

Founder Apr 2020 – Mar 2021

vyesna.com, Kathmandu, Nepal

- Built a Django/React job platform with a scraped-data recommender system.

Teaching & Volunteering

Teaching Assistant Dec 2024 – Jan 2025

NAAMII AI School, Kathmandu, Nepal

Lead Quantum Computing Researcher Mar 2021 – Mar 2023

OneQuantum, Munich (Remote)

Executive Member May 2023 – May 2024

Kathmandu University Robotics Club (KURC), Dhulikhel, Nepal

Data Scientist / Software Developer Mar 2021 – Mar 2022

Robotics Association of Nepal (RAN), Kathmandu, Nepal

Conference Talks & Panels

Guest Speaker, Career in AI/ML Dec 2023

CSIT Association of Nepal, Pokhara, Nepal

Conference Presentation, Nepali Stock Price Prediction Oct 2024

IEEE, Bhaktapur, Nepal

Conference Presentation, Landmark-Based Addressing Oct 2024

Kathmandu University, Dhulikhel, Nepal

Panelist, National Youth Scientific Conference 2024 Jun 2024

National Youth Council Nepal, Kathmandu, Nepal

Selected Projects

GreatAPI Jun 2022 – Present

Personal project, open-source Python web framework

Full-stack FastAPI framework; self-built and maintained

SahajMails Nov 2025 – Present

Personal project, open-source Python library

Personalized bulk email tool for outreach, newsletters, and campaigns; secure, cost-free email automation.

Solving Human Trafficking on a Quantum Computer

Nov 2021 – Apr 2022

Associated with OneQuantum

Applied the Quantum Approximate Optimization Algorithm and the Max-Cut problem to find optimal node configurations.

DistilBERT, Nepali News Classification

Aug 2022

Associated with IT MEET 2022

DistilBERT sequence-classification model over 9 Nepali newsgroup datasets; 95.475% accuracy.

Mission Oxygen Web Portal

May 2021 – Jun 2022

Associated with Robotics Association of Nepal

Facilitated access to vaccines, ventilators, oxygen cylinders, and hospital details during the COVID-19 pandemic.

Honors & Awards

Winner, Seeds for the Future Nepal, Huawei Technologies Co., Ltd.	Sep 2023
Winner, Infinity Codewave 2026 Hackathon, Kathmandu University	May 2026
Winner, Infinity Data Visualization Hackathon, Kathmandu University	May 2026
Winner, Computer Engineering Model Demonstration, Kantipur Engineering College	Dec 2024
Winner, KU Hackfest 2023 (OSM Category), Kathmandu University Computer Club	Oct 2023
Winner, Infinity Codewave, Department of Mathematics, Kathmandu University	Aug 2023
Winner, AI Competition (Text Challenge), IT MEET 2022, Kathmandu University	Aug 2022
Winner, AI Competition (Data Visualization), IT MEET 2022, Kathmandu University	Aug 2022
Winner, Youth Innovation Challenge 2022, Social Changemakers and Innovators	Aug 2022

Press & Media

Making Devanagari Handwriting Accessible: The Story of MallaNet, <i>Business 360°</i>	Feb 2026
Kathmandu University Student Builds an AI Model to Recognize Handwritten Devanagari Script, <i>Techpana</i>	Dec 2025
South Asia's 1st Quantum AI For Devanagari Digits – 99.8% Accuracy Breakthrough, <i>ICT Frame</i>	Jun 2025
Innovation on Home Soil: The Paths of Possibility Shown by Sahaj, <i>Terai Online</i>	Apr 2025
A Journey with “Seeds for the Future 2023” – A Technological Odyssey from Nepal to China, <i>ICT Frame</i>	Nov 2023

Technical Skills

Mathematics: Linear Algebra, Calculus, Statistics, Probability

Artificial Intelligence: PyTorch, LangGraph

Quantum Computing: Qiskit, PennyLane

Languages: Python, R, C++, TypeScript

Miscellaneous: Apache Spark, Apache Airflow, Docker, Amazon Web Services, Azure